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Standard
Recommended Practice

Liquid-Epoxy Coatings for External Repair, Rehabilitation, and Weld Joints on Buried Steel Pipelines

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Foreword

This standard recommended practice presents guidelines for establishing minimum requirements to ensure proper material selection, application, and inspection of pipeline liquid-epoxy coatings used for the repair and rehabilitation of previously coated pipelines and for coating field joints on the external surfaces of pipe. This standard is intended for use by corrosion control personnel, design engineers, project managers, purchasers, and construction engineers and managers.

Coating selection for repair, rehabilitation, and field joints on buried steel pipelines is determined by specific criteria relating to the project to be executed. In some cases, the factors limiting coating selection are similar to those considered during new pipeline coating activities, but when a pipeline is coated in the field there are additional factors that must be considered.

Liquid-epoxy coating repair, rehabilitation, or field joint coating can be applied in, over, or beside a ditch, whether the pipeline is in service, out of service, or segmented. Application is dependent on many factors, including temperature.

This standard was prepared by NACE Task Group 247 on Coatings, Liquid Epoxy for External Repair, Rehabilitations, and Weld Joints on Buried Steel Pipelines. This Task Group is administered by Specific Technology Group (STG) 03 on Coatings and Linings, Protective: Immersion and Buried Service. It is also sponsored by STG 04 on Coatings and Linings, Protective: Surface Preparation, and STG 35 on Pipelines, Tanks, and Well Casings. This standard is issued by NACE under the auspices of STG 03.

In NACE standards, the terms *shall*, *must*, *should*, and *may* are used in accordance with the definitions of these terms in the *NACE Publications Style Manual*, 4th ed., Paragraph 7.4.1.9. *Shall* and *must* are used to state mandatory requirements. The term *should* is used to state something considered good and is recommended but is not mandatory. The term *may* is used to state something considered optional.

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Contents

Section 1: General

1.1 This standard presents guidelines for establishing minimum requirements for material selection, surface preparation, proper application, inspection, and to ensure long-term coating performance of liquid-epoxy coatings applied to external pipe surfaces for coating repair, rehabilitation, or field joints.

1.2 The function of such coatings is to prevent corrosion when used in conjunction with cathodic protection.

1.3 This standard provides guidelines for proper surface preparation and coating application and identifies inspection and repair techniques to ensure long-term coating performance.

1.4 The material for external coating shall be approved by the owner. Once a coating formulation is accepted by the owner, it shall not be changed unless the owner is notified in writing.

1.5 The owner shall ensure that work is conducted in accordance with this standard and that quality is assured by qualified personnel such as NACE Coating Inspectors or equivalent (trained coating inspectors). The inspector shall have an excellent knowledge of the application procedures and techniques generally used for high-performance coatings whether applied by brush, roller, or spray. The coating inspector should know the characteristics of air- and airless-spray plural components, and shall be familiar with the procedures used to apply coatings properly with such equipment.

1.6 The coating inspector (company representative) shall have access at all times to the work performed, in accordance with the company specifications, and shall have the right to inspect such work and all material furnished by the applicator.

1.7 For surface preparation and coating applications the applicator shall use coating personnel, equipment, and procedures that have been prequalified by the manufacturer and approved by the owner.

1.8 Users of this standard shall be responsible for specifying applicable safety, health, and environmental practices to ensure compliance with all applicable regulations, including compliance with insurance requirements and directives.

Section 2: Definitions

Applicator: The organization responsible for the coating application.

Backfill: Material placed in a hole to fill the space around the anodes, vent pipe, and buried components of a cathodic protection system.

Backfill Ready: The stage or degree of cure, which may not be full chemical cure, that the coating has attained that provides resistance to moisture absorption and toughness to handle mechanical stresses including abrasion and backfill.

Batch: The quantity of coating material produced during a continuous production run of not more than eight hours (h).

Coating: A liquid, liquefiable, or mastic composition that, after application to a surface, is converted into a solid protective, decorative, or functional adherent film.

Cutback: The length of pipe left uncoated at each end for joining purposes (e.g., welding).

Epoxy: Type of resin formed by the reaction of aliphatic or aromatic polyols (like bisphenol) with epichlorohydrin and characterized by the presence of reactive oxirane end groups.

Holiday: A discontinuity in a protective coating that exposes unprotected surface to the environment.

Inspector: The authorized agent of the owner.

Owner: The owner company or the authorized agency that purchases the coating or coated pipe.

Supplier: The manufacturer and/or distributor of the coating material and its authorized technician.

Section 3: Coating Material

3.1 Coating Supplier Information

3.1.1 Upon request the coating material supplier shall furnish to the owner and/or applicator the following information in a written form:

3.1.1.1 Product data sheets including directions for handling, storage, and application of the coating material;

3.1.1.2 Specification of basic physical properties and laboratory performance test results as listed in Table 1.

3.1.1.3 Certification of the determined physical properties of supplied material; and

3.1.1.4 Material safety data sheet (MSDS).

TABLE 1
Coating Properties

Property	Value Limits	Test Method
Shelf Life	In accordance with supplier's specification; in accordance with supplier's recommended storage conditions	In accordance with supplier's specification
Nominal Coating Thickness (dry-film thickness [DFT] or wet-film thickness [WFT])	In accordance with supplier's specification	In accordance with supplier's specification
Cure Cycle (not applicable to all coatings)	In accordance with supplier's specification	In accordance with supplier's specification
Time to backfill-ready at 10ºC (50ºF), 24ºC (75ºF), 35ºC (95ºF), and 65ºC (150ºF)	In accordance with supplier's specification	In accordance with supplier's specification
Dielectric Strength	In accordance with supplier's specification	ASTM[(A)] D 149[1]
Adhesion to Steel	In accordance with supplier's specification	In accordance with supplier's specification
Moisture Permeation	Rating of 1 to 3	NACE Standard RP0394,[2] 28 days; tap water; at 24ºC (75ºF), 35ºC (95ºF), and 65ºC (150ºF)
Cathodic Disbondment	In accordance with supplier's specification, average of three specimens	NACE Standard RP0394[2], ASTM G 8,[3] ASTM G 42,[4] CSA[(B)] Z245.20,[5] 24 h and 28 days at 24ºC (75ºF), 35ºC (95ºF), 65ºC (150ºF), and at maximum recommended operating temperature in accordance with the manufacturer's tests
Temperature Usage Ranges	In accordance with supplier's specification	In accordance with supplier's specification
Impact Resistance	In accordance with supplier's specification	ASTM G 14[6]
Shore D Hardness	In accordance with supplier's specification	ASTM D 2240[7]
Cure time at 24ºC (75ºF), 35ºC (95ºF), and 65ºC (150ºF)	In accordance with supplier's specification	In accordance with supplier's specification

(A) ASTM International (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428.
(B) Canadian Standards Association (CSA), 5060 Spectrum Way, Mississauga, Ontario L4W 5N6 Canada.

3.2 Handling of Coating Materials

3.2.1 Coating material batches shall be identified by a batch coding system devised by the supplier. The batch code shall include a reference to the date of manufacture. Coating materials shall be shipped and stored under cover according to the supplier's recommendations and in such a manner that contamination or adverse effects on the coating material or application are avoided.

3.2.2 Shelf Life

The applicator shall ensure the material is within the shelf life recommended by the supplier. Any batch of coating material exceeding the supplier's recommended shelf life shall be subject to coating material reverification tests (see Table 1) prior to use.

3.3 Coating Material Properties

3.3.1 It is the supplier's responsibility to perform the tests cited in this section. The owner or applicator may also perform any or all of the cited tests as part of a quality assurance program.

3.3.2 The coating material shall meet the value limits for the properties listed in Table 1.

3.3.3 The owner may require and the supplier may provide information from tests other than those listed in Table 1.

Section 4: Coating Performance

4.1 Desired characteristics of applied coating are outlined in NACE Standard RP0169.[8]

4.2 Performance Testing

4.2.1 Qualification

The coating system should be qualified by the owner and/or coating applicator prior to coating application. This testing should be conducted on each applicable coating and meet the acceptance criteria (see Table 2). It is the responsibility of the owner to perform these tests. Once qualification is established, further qualification testing is not required unless there is a change in coating material, formulation, or location of manufacture. There may be a variety of other tests that the owner may wish to perform to verify the coating meets requirements.

4.2.2 Performance Testing Steel Panels

Test panels shall be hot-rolled carbon steel (UNS[(1)] G10200 [AISI[(2)] 1020] or equivalent) and have dimensions in accordance with the cited test method. The surface shall be blast cleaned to a near-white finish (NACE No. 2/SSPC[(3)]-SP 10[9]) using steel grit (G25, G40[10] 50 to 55 Rockwell C hardness [HRC]) to a surface profile of 50 to 100 µm (2 to 4 mil), measured from peak to valley.

4.2.3 Coating of Test Panels

4.2.3.1 Coating application and curing shall be in accordance with the supplier's recommendations.

4.2.3.2 Coating thickness on the test panel shall be measured using a magnetic pull-off coating thickness gauge calibrated against a NIST thickness standard (NIST[(4)]-SRM 1363[11]) that is within 20% of the specified coating thickness (SSPC-PA 2[12]) or magnetic-flux or fixed-probe gauges that must be calibrated with plastic shims on a representative grit-blasted surface.

4.2.4 Performance of Laboratory-Coated Steel Panels

4.2.4.1 Testing

A coating shall be considered qualified when the results of a duplicate set of test panels meet the acceptance criterion for each test shown in Table 2.

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(1) The UNS (Unified Numbering System) is a joint publication of the Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001; and ASTM International (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428.
(2) American Iron and Steel Institute (AISI), 1101 17th St. NW, Suite 1300, Washington, DC 20036.
(3) SSPC: The Society for Protective Coatings, 40 24th St., 6th Floor, Pittsburgh, PA 15222-4656.
(4) National Institute of Standards Technology (NIST), formerly the National Bureau of Standards, Gaithersburg, MD 20899.

TABLE 2
Qualification Requirements

Test	Acceptance Criteria	Number of Test Specimens	Test Method
Cathodic Disbondment (28 days) at 23ºC plus or minus 2ºC (68ºF)	Maximum radius: <10 mm (0.4 in.). No blisters.	3	NACE Standard RP0394,[2] CSA Z245.20,[5] ASTM G 8[3]
Cathodic Disbondment Maximum Operating Temperature. Required when the pipe is exposed at higher temperatures. The requirement applies only to coatings that may resist higher temperatures.	Maximum radius: <10 mm (0.4 in.) at 28 days, <25 mm (1.0 in.) at 90 days. No blisters.	3	ASTM G 42[4]
Impact Resistance	When using NACE Standard RP0394[2] or CSA Z245.20[5]—1.5 J (13 in.-lb) min. at 20ºC (68°F). No cracking or holidays. When using ASTM G 14[6]—3.5 J (30 in.-lb) min. at 24°C (75°F).	3	NACE Standard RP0394,[2] ASTM G 14,[6] CSA Z245.20[5]
Adhesion Test	When using NACE Standard RP0394[2]—a rating of 1 to 3 inclusive at 66°C (150°F). When using ASTM D 6677[13]—a rating of 6 to 10 at 20°C (68°F).	3	NACE Standard RP0394,[2] ASTM D 6677[13]
Penetration Test	10% of DFT at upper service temperature	3	ASTM G 17[14]

Section 5: Surface Preparation of Pipe

5.1 Prior to commencing surface preparation, the surface must be examined for the presence of any deleterious materials, such as heavy corrosion or rust scale, existing elastomeric or rubber-like coatings, or previously applied tape protective systems that require removal. If any such materials that cannot be readily removed by the blast cleaning process are present, preblasting surface preparation must be conducted. The preblasting surface preparation may use any company-approved processes, such as solvent cleaning (SSPC-SP 1[15]), use of hand or power tools (SSPC-SP 2[2] or SSPC-SP 3[16]), water blast cleaning or waterjetting (NACE No. 5/SSPC-SP 12[17]), or other removal methods. After removal, the surface should again be inspected to ensure that any deleterious material remaining may be readily removed during the subsequent blast cleaning operations.

5.2 This section emphasizes the importance of properly preparing the steel surface before the coating is applied. An uncontaminated and adequately profiled pipe surface shall be obtained because the condition of this interface directly affects the coating's performance.

5.2.1 Bare pipe shall be free of salts, mill lacquer, oil, grease, or other deleterious deposits that would prevent the coating from performing as expected. The applicator shall sufficiently remove any remaining deleterious deposits from the surface to be coated according to SSPC-SP 14. Any oil, grease, or magnetic particle inspection products, or ultrasonic couplant, shall be removed prior to blast cleaning, in accordance with SSPC-SP 1, using the company's approved solvent.

5.2.2 If moisture, frost, or dew is present on the steel surface, it shall be preheated to the temperature range recommended by the coating manufacturer to remove all moisture. Preheat shall be sufficient to ensure that the pipe temperature is at least 3°C (5°F) above the dew point temperature during cleaning, coating application, cure, and inspection.

5.2.2.1 This condition does not apply to surface-tolerant epoxies to be applied to wet or sweating surfaces. Special care should be taken when applying these products. The application of these products should be in accordance with the manufacturer's specifications.

5.2.2.2 NOTE: In-service pipelines with flowing product may cause the temperature of the pipe surface to be the same temperature as the product. If the product temperature is cold, external heat may not properly heat the surface to the temperature required for some coatings to be applied properly. Sweating could also present a problem. Proper hoarding, which consists of storing the working area inside a tent, and heaters are recommended when these conditions are present.

5.2.3 The exterior metal surface shall be abrasive blast cleaned to a near-white metal finish (NACE No. 2/SSPC-SP 10[9]) or better. A different surface finish should only be allowed when the procedure (e.g., thermite weld repairs or the use of surface-tolerant epoxies) has been prequalified by the manufacturer and approved by the owner.

5.2.4 Prior to coating, the cleaned pipe shall be visually inspected and imperfections (such as excessive weld cap or weld splatter) that may cause holidays in the coating shall be removed by grinding in such a manner as to give a surface finish suitable for coating application.

5.2.5 Pipe with imperfections that cannot be removed without encroaching upon minimum wall thickness shall be subject to replacement or repair.

5.2.6 When approved or specified by the owner, additional surface treatments may be used prior to application of coating. The applicator shall have available on site color prints of SSPC-VIS 1.[18] Material for abrasive cleaning shall be the appropriate blend of company-approved abrasive to produce an angular surface profile of 50 to100 µm (2 to 4 mil) measured from peak to valley in accordance with ASTM G 12.[19]

5.2.7 The edges of weld bevels and internal surfaces shall be protected during blast cleaning and coating application.

5.2.7.1 The blast media and coating material shall be prevented from entering the valves, fittings, and pipe. No amount of blast media in a valve, fitting, or pipe shall be acceptable.

5.2.8 Edges of the existing coating shall be roughened by power brushing or by sweep blasting but not removed. The coating shall overlap the existing coating for a minimum distance of 100 mm (4 in.).

5.2.9 The compressed air supply used for blast cleaning shall be free of water and oil. Separators and filters shall be used on the compressed air supply to ensure that contaminants such as oil and water are not deposited on the steel surface.

5.2.10 Residual blast products shall be removed from all blasted surfaces using a clean, dry, bristle brush; vacuum; or clean, dry compressed air.

5.2.11 Metal areas that develop flash rust because of exposure to moisture or humidity shall be given a sweep blast to return them to their original clean blasted condition. The surface to be blasted shall be recoated immediately. A sweep blast shall be carried out when a blasted surface is left overnight. The surface profile should remain in accordance with Paragraph 5.2.6.

Section 6: Coating Application

6.1 This section provides information on application techniques for obtaining optimum performance of each type of pipe coating discussed.

6.1.1 Surfaces to be coated shall be clean, dust-free, and dry, and shall meet all surface preparation requirements in Section 5 at the time of coating application.

6.1.2 Pipe and coating temperatures during application shall be in accordance with the coating supplier's recommendations.

6.1.2.1 Preheating procedures shall be sufficient to ensure complete coating cure during low ambient temperature applications. If preheating of the steel surface is required, it shall be accomplished without contaminating the steel surface.

6.1.3 The cure schedule shall be selected by the applicator to ensure adequate cure of the coating and should be in accordance with the coating manufacturer's recommendations.

6.1.4 The minimum coating thickness shall be specified by the owner. If there is concern about curing or other problems, maximum coating thickness may also be specified by the owner.

6.1.5 If preheating is required, the coating shall be applied immediately after preheating has been completed.

6.1.5.1 No solvents are to be added to the epoxy system.

6.1.6 The coating may be brush-grade or spray-grade as follows:

6.1.6.1 Brush-Grade: The coating is a two-component system (activator and base) and shall be mixed and applied in accordance with manufacturer's recommended practice.

6.1.6.2 Spray-Grade: The coating is a two-component system (activator and base) and shall be spray-applied in multiple passes to build the required thickness in accordance with the manufacturer's recommended practice.

6.1.7 During the coating application, the wet-film thickness (WFT) of the applied coating shall be measured using a wet-film gauge and the wet-film gauge marks shall be brushed out. Wet-film measurements shall be made on every joint, elbow, tee, etc., being coated. If low areas are detected, additional coating shall be applied before a tack-free condition occurs.

6.1.8 The coating shall cover the roughened existing coating at the overlap. (See Paragraph 5.2.8.)

6.1.9 The finished coating shall be generally smooth and free of application defects such as pinholes, fish eyes, sags, drips, icicles, etc.

6.1.10 After the coating has cured to a tack-free condition, the dry-film thickness (DFT) of each valve, fitting, and pipe section shall be measured using a magnetic gauge.

Section 7: Inspection and Testing

7.1 Tests Performed on Coated Pipe

7.1.1 The applicator shall have suitable inspection equipment at the construction site to perform tests as required in this section.

7.1.2 Verification of Coating Quality

7.1.2.1 The minimum testing frequency shall be established by the owner.

7.2 Inspection and Measurement

7.2.1 General

The inspection and measurement required in this section shall be made by the applicator under the supervision of a person qualified by experience and training in coating inspection methods. The results of these inspections shall be recorded and the results provided to the owner's representative.

7.2.2 Surface Preparation

The surface finish shall be monitored to determine compliance with Section 5 of this standard.

7.2.3 Surface Profile

The surface profile shall be measured using a replica tape or equivalent.[20] The profile shall be within the limits specified in Paragraph 5.2.6.

7.2.4 Inspection of Prepared Surface

Each cleaned pipe shall be visually inspected for surface defects and surface imperfections that may cause holidays in the coating. Surface imperfections shall be removed by grinding or other suitable means in accordance with Paragraph 5.1. Pipe containing surface defects shall be rejected or repaired at the owner's option.

7.2.5 Curing

7.2.5.1 The surface temperature of the pipe immediately prior to coating application shall be monitored and controlled within the limits recommended by the coating supplier. This process must be monitored and the values recorded at least once per hour or once per site if less than one hour.

7.2.5.2 In colder ambient conditions, especially in freezing or below freezing temperatures, preheating of the steel surface is required to cure the coating sufficiently before the pipe temperature returns to colder temperatures, which normally stops or retards the curing of epoxies. Proper preheating procedures should be applied according to the coating supplier specification.

7.2.5.3 The curing temperature and the time interval between application and backfill shall be controlled within the limits recommended by the coating supplier. This process must be monitored and the values recorded at least once per hour or once per site if less than one hour.

7.2.5.4 The methods that should be used to determine whether the coating is backfill-ready are adhesion tests such as the "cross hatch" and Shore D Hardness test.

7.2.5.4.1 Cross-Cut Test or Evaluating Adhesion by Knife in Accordance with ASTM D 6677.[13]

After the coating has cured, a "cross hatch" test shall be conducted using the procedure described in ASTM D 6677,[13] and the cross-hatch area should be repaired as mentioned in Section 8 after the test is complete.

Rating for the cross cut test:

- Refusal of the coating to peel or a cohesive failure within the coating shall be recorded as a "Pass." Cohesive failure is considered good.
- The extent of the adhesive failure between the coating and the metal substrate shall be recorded. Refer to ASTM D 6677[13] for the rating system.
- Cohesive failure, caused by voids leaving a honeycomb structure on the specimen surface, shall constitute a "Failure."
- Delamination of the coating shall also be considered a "Failure."

Cause for rejection:

- When the test fails to conform to the specified requirements, additional sampling shall be performed to ascertain the extent of failed coating. At the owner's option, all affected coating determined to have failed the testing may be rejected. Rejected coated pipe, valves, etc., shall be stripped and recoated.

7.2.5.4.2 Shore D Hardness Test

This test procedure involves obtaining coating samples approximately 760 µm (30 mil) thick from production runs cured under the same conditions and tested with a Shore D Hardness tester. This test may also be applied in the field by periodically measuring the hardness on the coated pipe. Coating is considered cured if the required Shore D Hardness reading, as listed by the coating manufacturer's specification, is obtained.

7.2.6 Coating Thickness

7.2.6.1 The coating thickness shall be measured in accordance with the owner's requirements. An example of a coating thickness requirement would be:

- Thickness measurements shall be taken using a coating thickness gauge calibrated at least once per eight-hour shift. All thickness measurements shall be recorded.
- If any individual measured thickness value is less than the specified minimum value of that coating, two coating thickness measurements shall be measured within 150 mm (6.0 in.) of the original measurement. The average of these measurements shall exceed the specified minimum value, and no individual value shall be more than 50 µm (2 mil) below the specified minimum value. All thickness measurements shall be recorded.
- Pipe failing to meet the above requirements shall, at the owner's option, be accepted, rejected, repaired, or stripped and recoated.
- If the owner's maximum acceptable thickness is exceeded, the coated pipe shall be further inspected, or stripped and recoated, as agreed on by the owner's representative and the applicator.

7.2.7 Holiday Inspection

7.2.7.1 General

7.2.7.1.1 The entire coated surface of each length of pipe shall be inspected with a holiday detector in accordance with NACE Standard RP0490.[21] The procedure shall be conducted in accordance with NACE Standard RP0490.[21] but with the minimum voltage set at 3.9 V/µm (100 V/mil.)

7.2.7.1.2 Coatings should not be inspected for holidays until after the coating is completely cured.

7.2.7.1.3 Pipe containing holidays shall be repaired or recoated in accordance with the requirements of Section 8.

Section 8: Repair

8.1 All coating defects should be repaired using owner-approved materials that are compatible with and adhere well to the base coating.

8.2 Repair materials normally consist of:

8.2.1 The same material used for the original repair, rehabilitation, or field joint coating; or

8.2.2 Repair materials and methods recommended by the coating supplier.

8.3 Repairs to isolated coating holidays should overlap the surrounding coating a minimum of 13 mm (0.50 in.).

8.4 The surface to be repaired should be clean, dry, and roughened with 80-grit sandpaper to ensure adhesion of the repair material.

8.5 Minimum thickness of the repaired coating shall be in accordance with the coating manufacturer's recommendations.

Section 9: Backfilling

9.1 After the pipe has been externally coated, the coating should be sufficiently cured to a backfill-ready state prior to backfilling.

9.2 Coated pipe shall be backfilled in such a way that damage to the pipe and/or coating is avoided.

References

1. ASTM D 149 (latest revision), "Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies" (West Conshohocken, PA: ASTM).

2. NACE Standard RP0394 (latest revision), "Application, Performance, and Quality Control of Plant-Applied, Fusion-Bonded Epoxy External Pipe Coating" (Houston, TX: NACE).

3. ASTM G 8 (latest revision), "Standard Test Methods for Cathodic Disbonding of Pipeline Coatings" (West Conshohocken, PA: ASTM).

4. ASTM G 42 (latest revision), "Standard Test Method for Cathodic Disbonding of Pipeline Coatings Subjected to Elevated Temperatures" (West Conshohocken, PA: ASTM).

5. CAN/CSA Z245.20-M86 M98, "External Fusion Bonded Epoxy Coated Steel Pipe" (Mississauga, Ontario, Canada: CSA).

6. ASTM G 14 (latest revision), "Standard Test Method for Impact Resistance of Pipeline Coatings (Falling Weight Test)" (West Conshohocken, PA: ASTM).

7. ASTM D 2240 (latest revision), "Standard Test Method for Rubber Property—Durometer Hardness" (West Conshohocken, PA: ASTM).

8. NACE Standard RP0169 (latest revision) "Control of External Corrosion on Underground or Submerged Metallic Piping Systems" (Houston, TX: NACE).

9. NACE No. 2/SSPC-SP 10 (latest revision), "Near-White Metal Blast Cleaning" (Houston, TX: NACE and Pittsburgh, PA: SSPC).

10. SAE[(5)] Standard J444 (latest revision), "Cast Shot and Grit Size Specifications for Peening and Cleaning, Recommended Practice" (Warrendale, PA: SAE).

11. NIST SRM 1363 (latest revision), "Non-Magnetic Coating on Steel," Certified Coating Thickness Calibration Standards (Gaithersburg, MD: NIST).

12. SSPC-SP 2 (latest revision), "Hand Tool Cleaning" (Pittsburgh, PA: SSPC).

13. ASTM D 6677 (latest revision), "Standard Test Method for Evaluating Adhesion by Knife" (West Conshohocken, PA: ASTM).

14. ASTM G17 (latest revision) "Standard Test Method for Penetration Resistance of Pipeline Coatings" (West Conshohocken) PA: ASTM.

15. SSPC-SP 1 (latest revision), "Solvent Cleaning" (Pittsburgh, PA: SSPC).

(5) Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001.

16. SSPC-SP 3 (latest revision), "Power Tool Cleaning" (Pittsburgh, PA: SSPC).

17. NACE No. 5/SSPC-SP 12 (latest revision), "Surface Preparation and Cleaning of Metals by Waterjetting Prior to Recoating" (Houston, TX: NACE and Pittsburgh, PA: SSPC).

18. SSPC-VIS 1 (latest revision), "Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning" (Pittsburgh, PA: SSPC).

19. ASTM G 12 (latest revision) 'Standard Test Method for Nondestructive Measurement of Film Thickness of Pipeline Coating on Steel" (West Conshohocken, PA: ASTM).

20. NACE Standard RP0287 (latest revision) "Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surface Using a Replica Tape" (Houston, TX: NACE).

21. NACE Standard RP0490 (latest revision) "Holiday Detection of Fusion-Bonded Epoxy External Pipeline Coatings of 250 to 760 µm (10 to 30 mils)" (Houston, TX: NACE).